

Surveying Apple Orchards with a Monocular Vision System

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Abstract—We present computer vision algorithms to collect yield related information in an apple orchard using images collected from a single camera. The goal of our system is to give farmers the capability to use their phones or digital cameras to record images and obtain yield related parameters. There are two challenges in this setup which necessitate novel methods: (i) It is very difficult to generate dense matches using standard image features. (ii) The constrained geometry of the setup causes existing structure from motion algorithms to fail. We present a novel structure from motion technique to register and reconstruct the apples which is used for extracting count and diameter information. We validate our approach by presenting results from multiple field trials.

I. INTRODUCTION

Modern precision agriculture techniques are revolutionizing farming by enabling farmers to collect, on demand, high-resolution data about the status of their crops. Access to such data helps in decision making for irrigation, fertilizer application, harvesting and sales. It also allows farmers to detect fruit diseases and nutrient deficiencies. In this work, we focus particularly on yield estimation in specialty farms such as apple orchards. We present a convenient method with which a farmer can collect images from a single camera (e.g. using a phone as shown in Figure 1) and obtain information regarding fruit count and size. Obtaining this information before harvest allows the farmer to make decisions regarding the number of pickers, the number of storage bins and make sales contracts early in the season. The information obtained also helps horticulture scientists by automating data collection e.g. for phenotyping.

Extracting yield related data is challenging. As a result, most existing methods operate under controlled lighting and imaging conditions (for example, they operate at night with artificial lighting). Others rely on stereo vision and active sensors such as laser scanners. Such systems, reviewed in Section II, are costly and not very convenient. The goal of the work presented in this paper is to provide farmers a mechanism to obtain yield-related data using only a single camera. Undoubtedly, cameras are the most widely used and one of the cheapest sensors available. Almost every phone today has a high resolution color camera good enough for data collection. We present computer vision techniques to extract yield and count information from such images.

Estimating Structure from Motion (SfM) is a fundamental problem in computer vision [1]. In a standard SfM pipeline



Fig. 1: An example usage of our system where a smartphone is used for collecting images. Our system extracts apple count and diameter information from such footage.

(Figure 3(a)) features such as Harris corners, SIFT [2] or SURF [3] are detected and matched across images. These matches are then used to estimate the geometry of the scene. Such methods fail in our application for two reasons: (i) *Features*: Often, point features such as SURF and SIFT are not detected on apples. Even when found, in the presence of occlusions and specularities these features can generate zero or false matches. Remaining correct matches are too sparse to estimate diameters. (ii) *Geometry*: baseline SfM methods fail to align dense apple clusters across multiple frames resulting in the same apple being registered multiple times (Figure 4).

Our primary contribution is a novel SfM pipeline where apple contours are directly used for generating dense point matches. These matches are then used for estimating the motion of the camera and reconstructing the apples in 3D. Our method is the first among such methods where fruit diameter information is obtained from images from a monocular camera. It should be noted that reconstruction from a single camera is bound to have global scale ambiguity (we can not know the scale without knowing the distance to the scene and vice versa). Our method fixes the reconstruction scale across the images, and normalizes the sizes with respect to a reference apple. This apple can be identified and measured by the user in order to obtain diameter information in a given unit (e.g. centimeters). We present results which validate our methods with footage obtained from a tablet and monocular cameras in multiple field trials.

We start with an overview of related work.

II. RELATED WORK

There has been significant recent activity for automating yield estimation [4]–[6]. Wang et al. [7] presented a system for apple yield estimation. Their system uses a side facing,

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wide-baseline vertical stereo rig mounted on an autonomous ground vehicle. It operates in controlled artificial illumination settings at night. In addition to count information, the stereo system extracts fruit sizes. Hung et al. [8] presented a feature based learning approach for identification of red and green apples and extraction of fruit count. Das et al. [9] developed a sensor suite for extracting plant morphology, canopy volume, leaf area index and fruit count that consists of a laser range scanner, multi-spectral cameras, a thermal imaging camera and navigational sensors. In contrast, our goal is to extract count and diameter from a single camera.

Jimenez et al. [10] present a survey of existing computer vision methods for locating fruit on trees. In particular, Mao et al. [11] developed methods for detecting Fuji apples based on color. However, these methods rely on images taken from a high-resolution camera mounted on a tripod and they only estimate yield in a single image. It is difficult to use such methods to obtain yield across the farm because multiple images can have overlapping fruit. In [12], a system for estimating the number of apples from color images in orchards under natural illumination was developed. This technique is also applicable on single images. The authors avoid specular lighting conditions by capturing images close to sunset. Our work does not rely on any particular assumption about the time of day or the position of the sun.

We have built a proof of concept system where a UAV equipped with a stereo pair captures images of apple trees, registers them in image space, and estimates the count and diameter of the apples [13]. In this paper, we remove the need for stereo. We first present an improved method for aligning apples across the images. Next, we present a new SfM pipeline to reconstruct the apples. This is then used to estimate diameters measured with respect to a reference apple.

III. PROBLEM FORMULATION AND ASSUMPTIONS

In developing our algorithms, we made a few assumptions about the setup. First, we assume that the camera is intrinsically calibrated. While auto-calibration techniques can be incorporated into the pipeline, we have not investigated this extension yet. It should also be noted that newer platforms such as Google Tango Tablet are making calibration parameters available to developers. As a result of the calibration assumption, we require the camera to have fixed focus. A camera whose focus changes dynamically (i.e auto-focus) is not compatible with our system. Most of the cameras in smart phones and tablets have fixed focal length and can be used for data collection (the auto-focus functionality in these cameras does not change the true focal length and camera center). Cameras with variable focal length can be used after fixing the zoom and focus. We have used the calibration toolbox and the standard chessboard pattern to calibrate the cameras [14].

We do not make any assumptions regarding environmental (e.g. illumination) conditions. The goal of the work presented in this paper can be summarized as: Given a sequence of images from a calibrated monocular camera, register all

apples, estimate their counts and find their diameters up to a user defined scale. As a by product, the observed apples are reconstructed up to a scale. For exact estimation of diameters, users can choose to scale the scene up or down according to metric measurements.

IV. OVERVIEW OF THE APPROACH

As mentioned earlier, we have previously developed a system for counting and diameter estimation using a stereo camera system. We utilize the preprocessing and post-processing components of this system which can be found in our technical report [13].

The primary components of our approach are:

- 1) Preprocessing involving the segmentation of input images to identify image contours containing apple clusters.
- 2) Apple clusters in consecutive segmented images are matched to compute dense features. This step is performed by computing an affine transformation to align two matching contours.
- 3) The dense features along with segmented images are then input to our SfM pipeline which registers the images and produces a reconstruction (up to a scale)
- 4) (Optional) The reconstruction is scaled by the user.
- 5) Post-processing: after reconstruction, the apples are counted and their diameters estimated.

Our contributions in this paper are the intermediate steps 2 and 3 which involve reconstruction using images from a single camera. We also investigate step 4 in section VIII.

V. COMPUTATION OF DENSE FEATURE MATCHES

In an orchard setting, usually only the frontal views of the apple clusters are available. Even from a few meters, the depth change in the visible portion of an apple is negligible and the apple geometry can be approximated by a circle lying on a plane. This allows us to make the assumption that matching apples across individual apple clusters in consecutive images are related by an affine transformation. In our earlier work [13], we have used exhaustive search to find the best affine transformation. In order to speed up process, we switched to an nonlinear optimization method based on Gauss-Newton gradient descent as described in [15]. In this section, we present the equations for apple contour alignment.

Given two apple contours, the affine transformation relating them is parameterized in the following way,

$$A(p) = \begin{bmatrix} 1 + p_1 & p_3 & p_5 \\ p_2 & 1 + p_4 & p_6 \end{bmatrix}$$

We seek the transformation A which minimizes

$$\sum_{\bar{x}} [I_2(\bar{x}) - I_1(A(\bar{x}))]^2 \quad (1)$$

where, I_1 and I_2 are the corresponding apple clusters and $\bar{x}$ is the homogeneous vector representation of image coordinates $[u, v, 1]^T$. Here, I_2 can be thought of as a template and I_1 as an image that is to be aligned to the template after

going through the affine transformation A . Note that we seek to compute a separate affine transformation for each apple cluster. The inverse compositional algorithm [15] seeks to minimize,

$$\operatorname{argmin}_{\Delta p} \sum_x [I_2(A(\Delta p)\bar{x}) - I_1(A(p)\bar{x})]^2 \quad (2)$$

here Δp is the incremental transformation to be applied to the existing affine warp $A(p)$. This warp is applied in the following way:

$$A(p) \leftarrow A(p) * A(\Delta p)^{-1}$$

As evident from the name, the incremental warp has to be inverted before composing with current warp. Performing a first order Taylor expansion of equation (2) we get,

$$\sum_{\bar{x}} \left[I_2(A(0)\bar{x}) + \Delta I_2 \frac{\delta A}{\delta p} - I_1(A(p)\bar{x}) \right]^2$$

If we assume that the initial warp $A(0)$ is identity, the incremental warp Δp can be computed as a least square solution to the above equation,

$$\Delta p = H^{-1} \sum_{\bar{x}} \left[\Delta I_2 \frac{\delta A}{\delta p} \right]^T [I_1(A(p)\bar{x}) - I_2(\bar{x})]$$

where the Hessian is

$$H = \sum_{\bar{x}} \left[\Delta I_2 \frac{\delta A}{\delta p} \right]^T \left[\Delta I_2 \frac{\delta A}{\delta p} \right]$$

The details of the algorithm can be found in the inverse compositional algorithm section in [15]. In step (2) of the inverse compositional algorithm [15] the error image is obtained as plain subtraction (i.e $I_1(A(p) * \bar{x}) - I_2(\bar{x})$). This step is changed in our implementation to penalize the warps that map the image points outside the template. We impose a heavy cost whenever, the warped points are outside the template.

Following the computation of dense matches, we can formulate an affine hierarchy between the matching contours across images and identify them uniquely. Figure 2(b) shows such an example. Affine transformation $A_{1,2}^1$ maps contour C_1 in image I_1 to I_2 and affine transformation $A_{2,3}^1$ maps contour C_1 from image I_2 to I_3 . Therefore the affine transformation relating contour C_1 from frame I_1 to frame I_3 is $A_{1,2}^1 * A_{2,3}^1$. We emphasize the novelty of our approach that different contours are related by different affine transformations (For contour C_2 we have a different affine hierarchy). The obtained dense matches and segmented images are next fed into our novel SfM pipeline, which is discussed in the next section.

VI. THE STRUCTURE FROM MOTION PIPELINE

Most of the standard feature based sequential SfM pipelines operate by selecting a pair of good initial frames for reconstruction (Figure 3(a)) [16]. After reconstructing the initial structure, each additional frame is matched to the frames already used in reconstruction. The reprojection error between reconstructed world points and matching points in the new frame is minimized to recover the camera pose and introduce new points to the structure [17]. Another standard approach, is to do two/three view reconstruction and then merging them using the constraints enforced by the common view [18]. Both these standard approaches work well for sparse reconstructions in the presence of large number of non-planar features. In contrast we have dense features from apples but many of them are planar. These standard approaches fail to align the reconstructed apples in different frames even with the aid of costly non-linear bundle adjustment (Fig. 4).

To resolve this problem, we propose a novel piecewise incremental structure from motion pipeline (Figure 3(b)). Similar to most incremental methods, our pipeline operates by pairwise reconstruction and merging the reconstructed point-clouds. As the cameras are calibrated, reconstructed points in different reference frames differ from each other by a similarity transformation. Our merging procedure utilizes the obtained affine relationship in the previous step to merge the reconstructions. Instead of solving for the transformation between two reference frames, we solve for the transformation between the matching apple clusters and align them individually.

Our novel pipeline for dense structure from motion has three steps:

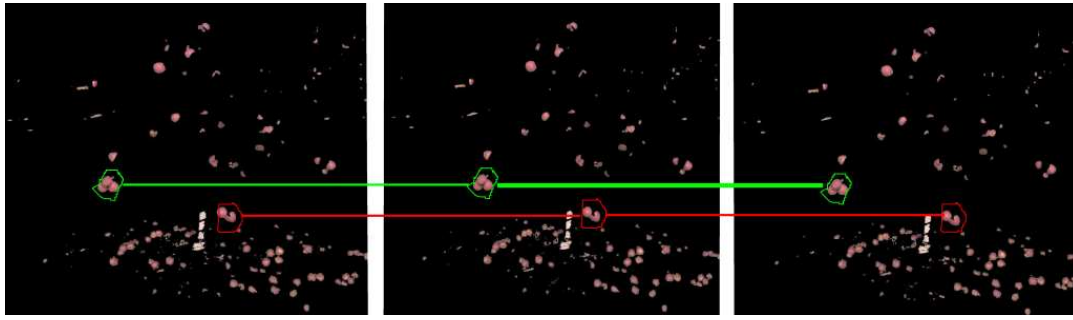
- 1) Keyframe selection and pairwise reconstruction
- 2) Alignment of reconstructed point-clouds using cluster-wise transformation
- 3) Computation of overall camera motion and optional bundle adjustment

We describe each of these steps in details in the following:

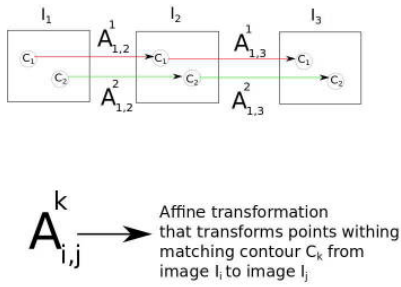
A. Keyframe Selection and Pairwise Reconstruction

To do precise reconstruction from an image sequence we need to ensure that frames within a pair satisfy the epipolar constraints, and have sufficient parallax between each other. Given the image sequence $I_1, I_2, \dots, I_n$ we choose an arbitrary frame I_m where $m \in \{1, 2, \dots, n\}$, as the first frame. Then we find another frame I_{m+i} which is consistent with frame m in terms of epipolar constraints. We perform pairwise reconstruction in the following way:

- 1) As the correspondence problem has already been solved in a previous step, we have dense feature matches between the images. To ensure enough parallax we compute both essential matrix E [19] and homography between the images and then check whether they satisfy the Geometric Robust Information Criterion (GRIC) defined by Torr [20]. This criterion



(a) Matching between contours.

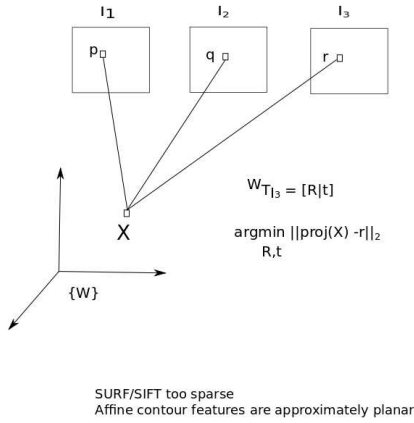


(b) Affine Map

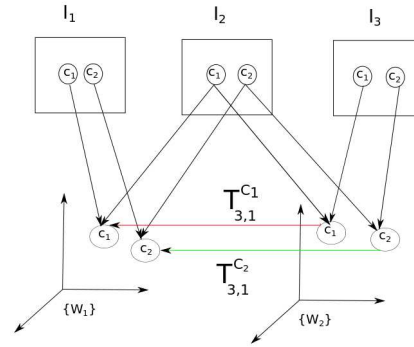


(c) Dense matching between images

Fig. 2: Computation of dense matching through affine transformation. For clarity only 10% of the matches are shown.



(a) A standard PnP based SfM pipeline



(b) Our piecewise incremental SfM pipeline

Fig. 3: Comparison of a common SfM pipeline and our pipeline. On Fig(a) we show a common structure from motion pipeline. It operates by constructing an initial structure. Then it incrementally augments the structure by adding new points from subsequent frames by solving the Perspective n Point (PnP) 2D to 3D correspondence problem. In the figure, matching points p, q and their reconstruction X is shown. The next camera pose is determined by minimizing the reprojection of the world point X to matching image point r . On Fig(b) we show Our SfM pipeline. Instead of aligning entire frames, we align the apple clusters individually. The average scale factor of these individual alignments are used to align non-matching clusters.

basically makes sure that the essential matrix is modelling the relationship between the frames better than homography. If the frames do not satisfy GRIC, we

skip the current frame and move to the next one.
 2) We decompose E to find the R and t between the cameras as described in section 9.6.2 of [21].

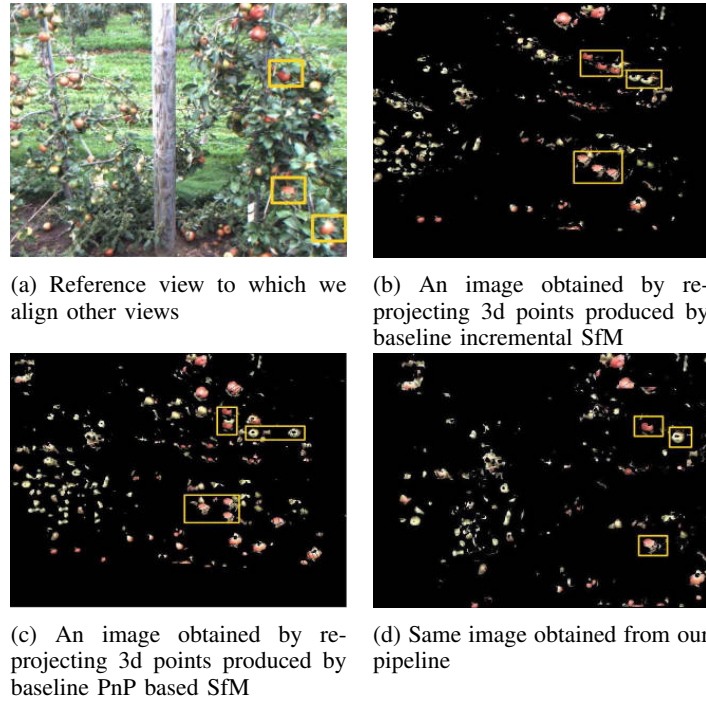


Fig. 4: Failure of baseline incremental SfM and PnP based methods to align dense apple clusters. Fig(a) shows the reference image to which we project all the 3D points from our reconstruction. Fig(b) shows a reprojected image frame from the baseline incremental SfM reconstruction and Fig(c) shows a reprojected image frame from a PnP based reconstruction. Both images contain same apples multiple times (some of them are shown in rectangular boxes) which shows the inability of these methods to register the clusters successfully. In Fig(d) we show the same image in Fig. 4 obtained from our pipeline. It solves the problem of multiple registration of the same apples observed while applying the baseline methods.

- 3) While decomposing E , if we find that, $\text{determinant}(E)$ is not close to zero (we use $\text{determinant}(E) < 1e-10$), or the two singular values (we use singular value ratio $\leq .9$) of E are not identical we reject the essential matrix and go to step 6.
- 4) Decomposition of E yields four combinations of R and t . We triangulate using each of these combinations and check for the chirality condition (we checked whether 99% of the reconstructed points are in front of the camera). If we find a valid triangulation with low average reprojection error (the threshold we use is 1.00 pixels), we reconstruct using the obtained projection matrices.
- 5) We impose one additional constraint on the triangulation angle to ensure the robustness of the reconstructed structure. Triangulated points for which the triangulation angle is within five to sixty degrees are kept and the rest are discarded. This is done to protect against parallel and vertical triangulation rays that lead to poor structure [22].
- 6) If we do not find a valid triangulation or we reject the essential matrix because of not satisfying the criteria in step 4, we discard the current frame and move to the next one.

Following these steps we reconstruct using the matches from the image pair, $\langle I_m, I_{m+i} \rangle$. We assume our

world/reference frame is the first camera frame. Next, we look for a subsequent frame which is consistent with frame I_{m+i} using the same procedure described above and reconstruct from the found pair. If none of the subsequent image frames are consistent with frame I_{m+i} , we discard one of the initial frames (e.g discard I_{m+i} and find a new suitable frame for pairwise reconstruction with frame I_m and continue in this way). As mentioned previously, the first camera frame used for reconstructing the initial pair acts as the frame of reference. Proceeding this way, we have multiple pairwise reconstructions, none of which are aligned. In the following section, we describe how to align these reconstructed point-clouds.

B. Alignment of reconstructed point-clouds using cluster-wise transformation

The alignment proceeds by calculating the transformation $T_{i,j}^{C_m}$ between matching reconstructed apple cluster C_m between reference frame i and j . Let iC_m denote the reconstructed cluster in reference frame i and jC_m denote the reconstructed cluster in reference frame j . Here, the reference frames also denote corresponding camera reference frames. The reconstruction in reference frame i is obtained from image pairs $\langle I_i, I_j \rangle$ and reconstruction in reference frame j is obtained from pair $\langle I_j, I_k \rangle$. So,

$${}^iC_m = {}^iR_j^jC_m + \lambda_i^jT_{j \rightarrow i}^{org} \quad (3)$$

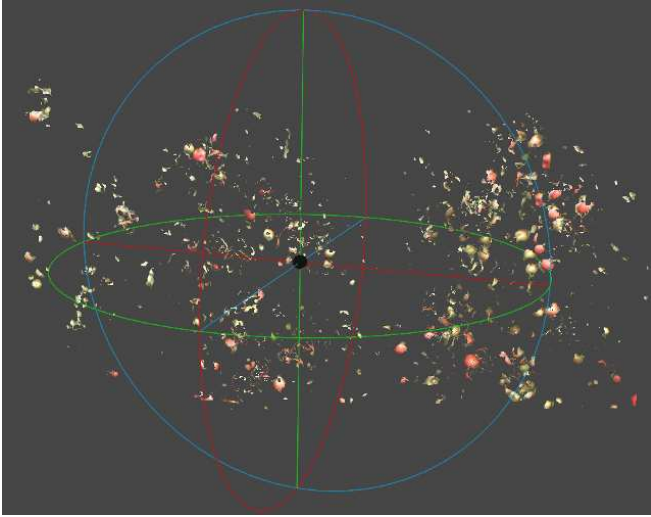


Fig. 5: Dense reconstruction from our pipeline. We show a sample reconstruction of three trees from our novel SfM pipeline. The circles are shown for illustration purposes.

where ${}^iR_j, {}^iT_{j_{org}}$ is the rotation and translation mapping reference frame j to i , λ_i is the lost scale factor while extraction of translation from essential matrix. If i is our base reference frame (i.e all other reconstructions will be mapped to this frame), without loss of generality we can set $\lambda_i = 1$. All other frames will be reconstructed up to the scale of the initial reconstruction in base frame i . Moving onto the next reconstruction of cluster C_m in reference frame j ,

$${}^jC_m = {}^jR_k^k C_m + \lambda_j^j T_{k_{org}} \quad (4)$$

Replacing the value of jC_m in equation (3) from equation (4) we obtain:

$${}^iC_m = {}^iR_j ({}^jR_k^k C_m + \lambda_j^j T_{k_{org}}) + \lambda_i^i T_{j_{org}} \quad (5)$$

Using the correspondences from the cluster C_m in equation (5) we can solve for the scale factor λ_j . Then the points in reconstruction of jC_m in reference frame j which do not correspond to any points in iC_m can be aligned to base frame i using the right side of equation (5). An illustration of this procedure can be found in Figure 3(b). As equation (5) uses only correspondences within a cluster C_m and therefore the obtained scale is local with respect to the reconstructed points in iC_m . If we average all of these local scales, we will obtain a scale consistent across the entire point-cloud. This scale is used to align point-clouds of clusters that do not have corresponding matching clusters in the previous reconstruction.

C. Computation of overall camera motion and optional bundle adjustment

The retained overall scale in the previous step defines the additional camera pose up to a consistent scale. The pose for camera k is, ${}^jT_k = \begin{bmatrix} {}^jR_k | \lambda_G^j T_{k_{org}} \end{bmatrix}$, where λ_G denote the overall scale. In conventional incremental SfM, this step is followed by a bundle adjustment [23]. The robustness of

the piecewise alignment enables us to skip this step. Figure 8 shows, even without bundle adjustment the reprojection error is much lower than the baseline incremental and PnP based methods. Nevertheless, we perform bundle adjustment as the final step of the reconstruction procedure. It helps to remove small drifts in the camera motion resulting in robust estimation of both structure and motion (Fig. 7).

VII. COUNTING AND DIAMETER ESTIMATION

Once we obtain the full structure from all the frames, we are in a good position to estimate the count and diameter of the apples.

For counting the number of apples, the counting method in [13] is used. The method takes registered segmented apple images as input. As we have already registered the apples in 3D space, it is trivial to generate the 2D registered images. We divide our total structure into arbitrary portions and project the 3D point-clouds of apple clusters to a camera frame where they are visible.

As apple clusters are tricky to separate into individual apples, we select only single apples for diameter estimation. This is ensured by the counting method. To compute the diameters, we use the diameter estimation method in [13]. This method essentially finds the pixel diameter and average depth of the pixels within the apple. Then it projects the pixel diameter to average depth to find the metric diameter. As our reconstruction is only up to a scale, the obtained diameter is also defined upto a scale. We fix the diameter of a random apple in metric unit to obtain the required scale factor. This scale factor is used to convert our obtained up to scale diameters to metric units. The metric diameter of the random apple is a user input in our estimation system. In the section, VIII-B we evaluate our counting and diameter estimation procedures rigorously.

VIII. EXPERIMENTAL RESULTS

As outlined previously our main contribution in this paper, is a novel sequential structure from motion pipeline for reconstruction in orchard settings. Utilizing, this pipeline we obtain a dense reconstruction of orchard rows and estimate fruit counts and diameters. We collected many different data sets using stereo (for ground truth of diameter data) and monocular cameras. In this paper we have used two datasets:

- Dataset1 primarily contains red Apples. Total number of images are 965. Sample images from the dataset are shown in Fig. 6. These images covers a full row of an orchard.
- Dataset2 contains a mixture of red and green apples. Total number of images are 464. Sample images from the dataset are shown in Fig. 6. This dataset covers one block containing six trees.

We tested both our SfM pipeline and yield estimation capabilities using these two data sets.

A. Evaluation of the novel SfM pipeline

Our SfM pipeline uses a piecewise incremental process to recover camera motion and scene structure. The figures



Fig. 6: Sample images from used datasets. The figure on left shows a sample image from Dataset1 primarily containing red apples. The figure on right shows a sample image from Dataset2 containing a mixture of red and green apples.

(Fig. 7, 8) are produced from over 200 frames in Dataset2. The figures show only the keyframes selected by our SfM pipeline. Even without the aid of bundle adjustment the reprojection error (Fig. 8) is very low. However, the camera motion without bundle adjustments have small drifts as shown in Fig. 7. The bundle adjustment procedure eliminates these small drifts and corrects the camera motion and scene structure.

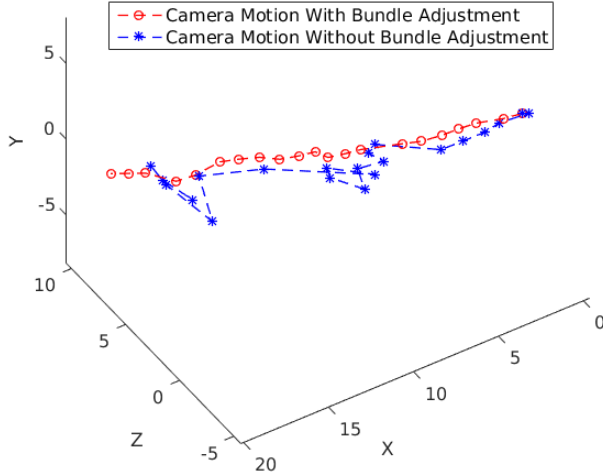


Fig. 7: The computed camera motion with and without bundle adjustment. Bundle adjustment removes the drifts in camera motion.

We compare our method with the two popular baseline methods: incremental SfM and PnP based SfM. Both methods are implemented with full bundle adjustment after addition of each new frame. The results show that even without bundle adjustment our piecewise incremental method has far lower reprojection error than the baseline methods (Fig. 8).

B. Field results for apple count and diameter estimation

As discussed earlier, the full description of the counting and diameter estimation algorithm is beyond the scope of this paper. For details of these methods please see [13]. Field

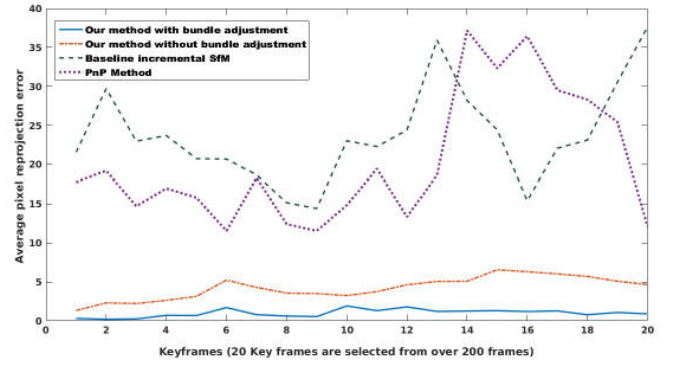


Fig. 8: Reprojection error comparison. Our piecewise incremental method performs better than both baseline methods even without bundle adjustment. Bundle adjustment though reduces the reprojection error further and helps to correct the small drifts and smooth camera motion as shown in Fig. 7.

trials have shown that the counting accuracy for Dataset1 (containing red apples only) and Dataset2 (containing both red and green apples) is 85.87% and 81.3492%.

For evaluating diameter estimation using monocular cameras, we use stereo data as ground truth. The diameters from our SfM reconstruction is only up to a scale. To fix the scale, we randomly choose an apple and find the metric value of its diameter. The ratio of the obtained diameter from our pipeline and this metric value provides us with the scale factor needed to scale our diameter data. Fig. 9 shows the histogram of obtained diameters from stereo and single camera. We compared the diameters of hundred single apples from the Datasets. The results show that the mean difference between monocular and stereo diameter is $+0.90$ centimeters(cm) and the standard deviation is ± 2.9 cm. To find the dissimilarity between the two distributions more rigorously, we used earth mover's distance [24]. The found dissimilarity between the two distribution is .65. In other words, to morph one distribution into another, the average distance per weight in the horizontal axis is less than one.

Using only one apple to scale up the scene can be susceptible to noise. To get accurate estimates we may need to choose multiple apples, find their actual diameter and find

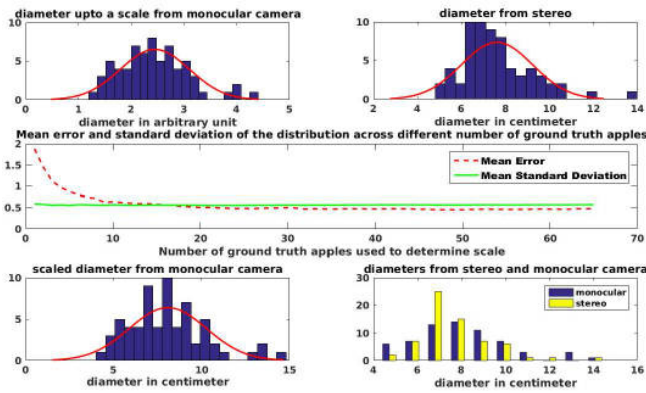


Fig. 9: Comparison of diameter estimation from Monocular Camera (top-left) with Stereo (top-right). Middle row: The scale can be estimated by measuring apples as ground truth. The error as a function number of apples used to estimate the scale. With four or more measurements, the error drops below one centimeter. Bottom row: The scaled diameter estimation from our method (left) and a side by side comparison of obtained histograms (right).

the average scale factor. Figure 9 shows how the error in the monocular estimate decreases with increasing number of apples used for estimating scale. The results indicate that if we use more than four ground truth apples, the error drops below one cm.

IX. CONCLUSION

This paper presented first results for a system which can produce yield estimates from monocular images. The system provides count information and a distribution of the relative size. If the absolute size is required, the only input required from the farmer would be the true size of a single measured apple. From a technical viewpoint, our main contribution is a method to reconstruct the structure of the fruit by generating dense matches from contour correspondences. The techniques described in this paper require cameras to be calibrated. They are also executed offline after data collection. In future, we would like to extend these techniques for non-calibrated cameras. We are also interested in real time implementations (using GPU available on smartphones) so that the application runs directly on the platform while data is being collected.

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